Spark Tamers — Technical Sheet

Tech used

Single self-contained HTML file (about 44 KB), no libraries, images or external assets. HTML5 Canvas 2D with devicePixelRatio scaling (capped at 2x).

Graphics: all 15 creatures are drawn in code with type-specific features (flames, fins, leaves, lightning horns, rock spikes, orbiting wisps), radial-gradient glowing bodies, blinking eyes, idle bobbing, and bigger, angrier evolved forms. Swaying neon grass with fireflies, rippling water, glowing shrine and shop tiles, pulsing Warden gates, a follower creature, battle arenas with skies per zone, hit flashes, shakes, type-colored particle bursts, an orb throw and wobble, smooth HP bars and an encounter wipe transition.

Web Audio API: all sound is synthesized: steps, grass rustle, encounter jingle, hits (heavier when super effective), misses, buffs, heals, orb throw and wobble, catch fanfare, level-up, evolution, victory. Procedural music: a calm major-key overworld theme and a faster minor-key battle theme with drums. Audio starts on first tap (iOS-safe), can be muted, and suspends when the tab is hidden.

Pointer Events floating joystick, 56-64px buttons, safe-area insets, portrait layout. Saves to localStorage.

Game systems

15 species in 6 types with a full type chart, 22 moves (attack, buff and heal), level-based stats, XP curve, move learning (weakest move is replaced), evolution, catch formula based on HP and rarity with 3 orb grades.

Map of 4 zones (15x26 tiles each) generated from a fixed seed: tall grass, trees, water, a winding path, a shrine and shop per zone and a Warden gate. Each zone has its own wild pool and level range.

4 Wardens with teams of 2 to 4 creatures (levels 8 to 26). Opponent AI picks moves by expected damage, finishes off low foes, heals when low and buffs early; Warden AI plays without randomness.

Audit 1 — Balance (simulated play)

A headless simulator uses the same game code. Three bot styles each played 30 full campaigns from a random starter: a rush bot that challenges Wardens under-leveled and never catches, a casual bot that picks moves loosely and challenges at the Warden’s top level, and a smart bot that catches a team and levels 2 above.

First pass: the XP curve was too slow. Bots needed about 220 wild battles to finish, which is too grindy on a phone. Raised XP per win by 50% and flattened the level curve.

Final results: rush: 19/30 champions, avg 87 wild battles, avg tries per warden 3.1/3.3/4.6/5.7; casual: 30/30 champions, avg 88 wild battles, avg tries per warden 1.1/1.3/1.5/2.4; smart: 30/30 champions, avg 112 wild battles, avg tries per warden 1.0/1.0/1.1/2.0. Players who catch a team and use type advantages finish reliably; rushing under-leveled usually fails at the last Wardens.

Audit 2 — Bugs

Ran a headless smoke test of the real file with a stubbed browser: title, starter pick, 120 wild battles with fighting and catching, forced swaps after fainting, evolution, Warden 1, passing the gate, the shop, the team screen, mute and saving. No errors.

Fixed: the encounter wipe was tied to the global clock, so it could appear already half done. It now starts from the moment of the encounter, and the walking joystick is released so you don’t keep walking after a battle.

Fixed: saves stored temporary battle data (stat boosts and HP bar animation). It is now stripped, so boosts can’t carry over after reloading.

Audit 3 — Player experience

A first-steps hint (drag to walk) and an intro toast explain grass, shrine, shop and gates.

Move buttons show type, power and whether a move is STRONG, weak or has no effect against the current foe. The Orb button shows the exact catch chance.

Fainting is gentle: you go back to the shrine and keep everything except a quarter of your gold. Locked gates explain who blocks them, and the shop only shows better orbs once they matter.